सी एस आई आर – राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद)
(Council of Scientific and Industrial Research)
(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
(National Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
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सीएसआईआर CSIR भारत का नवाचार इंजन The Innovation Engine of India

अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25050115/D1.01/C-028

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
19.06.2025	19.06.2028	1	2

1. **Calibrated for** : Zenith Calibration & Research Center
   Kathmandu, Nepal

   Customer's Ref. No. : Letter Dated : 22.04.2025
   From : Harriss & Tarriss, Haryana

2. **Description and identification of instrument**

1 kg to 1g (13 Nos.)	:	Integral Knob cylindrical stainless steel weights.
Assumed Density (d)	:	(7950 ± 140) kg/m^3; $k = 2$
Make	:	Tycon
Model No.	:	TYCO-01 (for 200 g to 1 g)
Serial No.	:	202300014 (for 200 g to 1 g)

3. **Environmental conditions** : Temperature : (23.0 ± 1.5) °C
   Relative humidity : (50 ± 10) %
   [Change in temperature during the calibration was less than ± 0.7 °C per hour]

4. **Standard(s) used with associated Measurement uncertainty** : NPL working standard (Sub-Div.#1.01/M/224(2)/00) of mass with uncertainty better than one-third of the reported uncertainty of measurement.

5. **Metrological Traceability of Standard(s) used** : The standard(s) used for calibration is(are) traceable to the National Standard which realize the units of quantities according to the International System of Units (SI).

6. **Principle/Methodology of Calibration & Calibration Procedure number** : CSIR-NPL Calibration Procedure No. Sub-Div.#1.01/Doc.3/ CP#WT/M-03. Method of comparison with the CSIR-NPL working standard(s) using substitution weighing. The reported mass value(s) is(are) the conventional mass value(s) (M_c) related to the true mass value(s) (M_t) by formula : $M_c = M_t \times [1 - 1.2 \times (1/d - 1/8\,000)]$.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by:
National Physical Laboratory Director's Nominee
डा. श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam

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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25050115/D1.01/C-028

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pa ges
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7. Results :

Denomination	Mass Value (g)	Uncertainty (g)
1 kg	999.999 7	± 0.000 5
500 g	499.999 62	± 0.000 25
200 g	200.000 12	± 0.000 10
. 200 g	200.000 02	± 0.000 10
100 g	100.000 10	± 0.000 05
50 g	50.000 06	± 0.000 03
20 g	20.000 007	± 0.000 025
. 20 g	20.000 013	± 0.000 025
10 g	9.999 993	± 0.000 020
5 g	5.000 024	± 0.000 016
2 g	1.999 992	± 0.000 012
. 2 g	2.000 002	± 0.000 012
1 g	1.000 010	± 0.000 010

The reported uncertainty is at coverage factor $k = 2$ which corresponds to a coverage probability of approximately 95 % fo a normal distribution.

8. Date(s) of Calibration : 06.06.2025

9. Remarks :
1. Mass value(s) of the weight(s) is(are) within the maximum errors permissible ir E_2 accuracy class of weights as per OIML R 111-1 : 2004.
2. Dots(.) are used to distinguish the weights of same nominal value.

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प्रभारी वैज्ञानिक:
Scientist-in-charge:
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Director's Nominee
डॉ. श्रीनिवास राव रागम
Dr. Srinivas Rao Ragam